Angle in a Semicircle

Exam code: 9MA0

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Written by: Roger B
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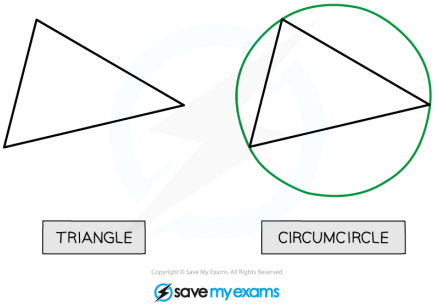
Angle in a semicircle

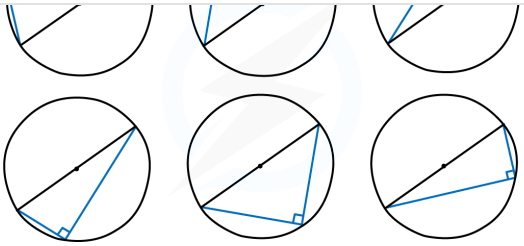
What are angles in a semicircle?

- It is always possible to draw a unique circle through the three vertices of a triangle – this is called the **circumcircle** of the triangle



- The **angle in a semicircle property** says that if a triangle is right-angled, then its hypotenuse is a diameter of its circumcircle

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



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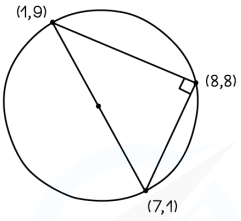
- It also says that **any** angle at the circumference in a semicircle is a right angle

How can I use angles in a semicircle to find the equation of a circle?

- Because the hypotenuse of a right-angled triangle is a diameter of the triangle's circumcircle you also know that:
  1. the radius of the circumcircle is half the length of the hypotenuse
  2. the centre of the circumcircle is the midpoint of the hypotenuse



CENTRE OF CIRCLE = MIDPOINT OF HYPOTENUSE

$= \left(\frac{1+7}{2}, \frac{9+1}{2}\right)$

$= (4,5)$

MIDPOINT OF LINE SEGMENT FROM (x_1, y_1) TO (x_2, y_2) IS $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

RADIUS $= \frac{1}{2} \times$ LENGTH OF HYPOTENUSE

$= \frac{1}{2}\left(\sqrt{(7-1)^2+(1-9)^2}\right)$

$= \frac{1}{2}\sqrt{100} = 5$

LENGTH OF LINE SEGMENT FROM (x_1, y_1) TO (x_2, y_2) IS $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

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- Once you know the radius and the centre you can write down the **equation of the circle**

EQUATION OF CIRCLE WITH CENTRE (4,5) AND RADIUS OF 5 IS

$(x-4)^2+(y-5)^2=5^2$

$(x-4)^2+(y-5)^2=25$

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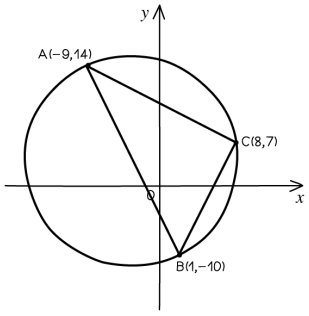
Examiner Tips and Tricks

- To show that a triangle is right-angled, show that the lengths of its sides satisfy Pythagoras' theorem.

Worked Example

The three points $A\ (-9,\ 14)$, $B\ (1,\ -10)$ and $C\ (8,\ 7)$ lie on a circle, as shown in the diagram.



a) Show that AB is a diameter of the circle

b) Find an equation of the circle

a) LENGTHS OF THE SIDES ARE:

$AB^2 = (1-(-9))^2 + (-10-14)^2 = 676$

$AC^2 = (8-(-9))^2 + ((7-14))^2 = 338$

$BC^2 = (8-1)^2 + (7-(-10))^2 = 338$

BECAUSE WE'RE DEALING WITH PYTHAGORAS, IT'S EASIER TO KEEP EVERYTHING SQUARED HERE

$AC^2 + BC^2 = 338 + 338 = 676 = AB^2$

∴ BY PYTHAGORAS ABC IS A RIGHT-ANGLED TRIANGLE, AND AB IS THE HYPOTENUSE

∴ BY THE ANGLE IN A SEMICIRCLE PROPERTY, AB IS THE DIAMETER OF THE CIRCLE THROUGH A, B AND C

b) LENGTH OF AB $= \sqrt{676} = 26$

RADIUS OF CIRCLE $= \frac{1}{2} \times 26 = 13$

CENTRE OF CIRCLE = MIDPOINT OF AB

$= \left(\frac{-9+1}{2}, \frac{14+(-10)}{2}\right)$

$= (-4, 2)$

EQUATION OF THE CIRCLE IS

$(x-(-4))^2 + (y-2)^2 = 13^2$

$(x+4)^2 + (y-2)^2 = 169$

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